

Large Dimensional Regressions

M514: Econometrics II

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Naive Shrinkage Estimator

Shrinkage Estimator

Suppose we compare a shrinkage estimator of the form

$$\tilde{\beta} = \frac{1}{1 + \lambda} \hat{\beta}$$

where $\hat{\beta}$ is the OLS estimator. The bias of $\tilde{\beta}$ is given by

$$\begin{aligned}\mathbb{E}[\tilde{\beta}|\mathbf{X}] - \beta &= \mathbb{E}\left[\frac{1}{1 + \lambda} \hat{\beta}|\mathbf{X}\right] - \beta = \frac{1}{1 + \lambda} \beta - \beta \\ &= -\frac{\lambda}{1 + \lambda} \beta\end{aligned}$$

and the variance of $\tilde{\beta}$ is given by

$$\begin{aligned}\mathbb{E}\left[\left(\tilde{\beta} - \mathbb{E}[\tilde{\beta}|\mathbf{X}]\right) \left(\tilde{\beta} - \mathbb{E}[\tilde{\beta}|\mathbf{X}]\right)' | \mathbf{X}\right] &= \left(\frac{1}{1 + \lambda}\right)^2 \mathbb{E}\left[\left(\hat{\beta} - \mathbb{E}[\hat{\beta}|\mathbf{X}]\right) \left(\hat{\beta} - \mathbb{E}[\hat{\beta}|\mathbf{X}]\right)' | \mathbf{X}\right] \\ &= \left(\frac{1}{1 + \lambda}\right)^2 V\end{aligned}$$

wMSE of the Shrinkage Estimator

Recall that the weighted MSE can be written as

$$\text{wMSE}(\tilde{\beta}|\mathbf{X}) = \left(\mathbb{E}[\tilde{\beta}|\mathbf{X}] - \beta\right)' W \left(\mathbb{E}[\tilde{\beta}|\mathbf{X}] - \beta\right) + \mathbb{E} \left[\left(\tilde{\beta} - \mathbb{E}[\tilde{\beta}|\mathbf{X}]\right)' W \left(\tilde{\beta} - \mathbb{E}[\tilde{\beta}|\mathbf{X}]\right) | \mathbf{X} \right]$$

Let's characterize how the weighted bias and the weighted variance of $\tilde{\beta}$ behave as we change λ .

Weighted Bias squared

The weighted bias squared is given by

$$\left(\mathbb{E}[\tilde{\beta}|\mathbf{X}] - \beta\right)' W \left(\mathbb{E}[\tilde{\beta}|\mathbf{X}] - \beta\right) = \frac{\lambda^2}{(1 + \lambda)^2} \beta' W \beta$$

because the bias of $\tilde{\beta}$ is given by

$$\begin{aligned}\mathbb{E}[\tilde{\beta}|\mathbf{X}] - \beta &= \mathbb{E}\left[\frac{1}{1 + \lambda} \hat{\beta}|\mathbf{X}\right] - \beta = \frac{1}{1 + \lambda} \beta - \beta \\ &= -\frac{\lambda}{1 + \lambda} \beta\end{aligned}$$

Weighted Variance

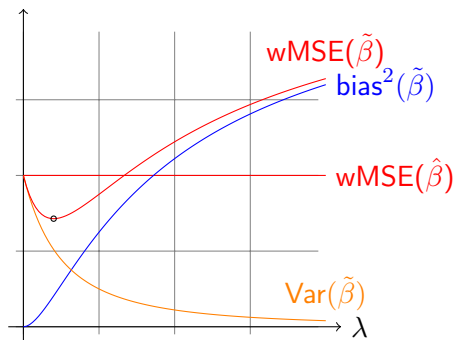
The weighted variance is given by

$$\begin{aligned}\mathbb{E} \left[\left(\tilde{\beta} - \mathbb{E}[\tilde{\beta}|\mathbf{X}] \right)' W \left(\tilde{\beta} - \mathbb{E}[\tilde{\beta}|\mathbf{X}] \right) | \mathbf{X} \right] &= \text{tr} \left(\mathbb{E} \left[\left(\tilde{\beta} - \mathbb{E}[\tilde{\beta}|\mathbf{X}] \right)' W \left(\tilde{\beta} - \mathbb{E}[\tilde{\beta}|\mathbf{X}] \right) | \mathbf{X} \right] \right) \\&= \mathbb{E} \left[\text{tr} \left(\left(\tilde{\beta} - \mathbb{E}[\tilde{\beta}|\mathbf{X}] \right)' W \left(\tilde{\beta} - \mathbb{E}[\tilde{\beta}|\mathbf{X}] \right) \right) | \mathbf{X} \right] \\&= \mathbb{E} \left[\text{tr} \left(W \left(\tilde{\beta} - \mathbb{E}[\tilde{\beta}|\mathbf{X}] \right) \left(\tilde{\beta} - \mathbb{E}[\tilde{\beta}|\mathbf{X}] \right)' \right) | \mathbf{X} \right] \\&= \text{tr} \left(\mathbb{E} \left[W \left(\tilde{\beta} - \mathbb{E}[\tilde{\beta}|\mathbf{X}] \right) \left(\tilde{\beta} - \mathbb{E}[\tilde{\beta}|\mathbf{X}] \right)' \right) | \mathbf{X} \right] \right) \\&= \text{tr} \left(W \mathbb{E} \left[\left(\tilde{\beta} - \mathbb{E}[\tilde{\beta}|\mathbf{X}] \right) \left(\tilde{\beta} - \mathbb{E}[\tilde{\beta}|\mathbf{X}] \right)' \right) | \mathbf{X} \right] \right) \\&= \left(\frac{1}{1 + \lambda} \right)^2 k\end{aligned}$$

because we use $W = V^{-1} = \left(\mathbb{E} \left[\left(\tilde{\beta} - \mathbb{E}[\tilde{\beta}|\mathbf{X}] \right) \left(\tilde{\beta} - \mathbb{E}[\tilde{\beta}|\mathbf{X}] \right)' \right) \right)^{-1}$.

Graph of wMSE of $\hat{\beta}$, $\tilde{\beta}$

$$\begin{aligned} \left(\mathbb{E}[\tilde{\beta}|\mathbf{X}] - \beta \right)' W \left(\mathbb{E}[\tilde{\beta}|\mathbf{X}] - \beta \right) &= \frac{\lambda^2}{(1+\lambda)^2} \beta' W \beta \\ \mathbb{E} \left[\left(\tilde{\beta} - \mathbb{E}[\tilde{\beta}|\mathbf{X}] \right)' W \left(\tilde{\beta} - \mathbb{E}[\tilde{\beta}|\mathbf{X}] \right) | \mathbf{X} \right] &= \left(\frac{1}{1+\lambda} \right)^2 k \end{aligned}$$



You can always do better than $\hat{\beta}$ by shrinking it by a small amount.

wMSE of $\tilde{\beta}$

Given the weighted MSE for $\tilde{\beta}$,

$$\text{wMSE}(\tilde{\beta}|\mathbf{X}) = \frac{\lambda^2}{(1 + \lambda)^2} \beta' W \beta + \frac{1}{(1 + \lambda)^2} k$$

the value of λ which minimizes the wMSE of $\tilde{\beta}$ is given by

$$\lambda = \frac{k}{\beta' W \beta}$$

The minimized value is given by $\frac{1}{\frac{1}{k} + \frac{1}{\beta' W \beta}}$. We see that only in the extreme case where $\beta' W \beta = \infty$ do we have that the wMSE of $\hat{\beta}$ is equal to the wMSE of $\tilde{\beta}$ with optimally chosen λ .

Choice of λ

How do we interpret $\beta'W\beta$?

Suppose that conditional homoskedasticity holds, i.e., $V = \sigma^2(\mathbf{X}'\mathbf{X})^{-1}$.

$$\begin{aligned}\beta'W\beta &= \beta'V^{-1}\beta \\ &= \sigma^{-2}(\mathbf{X}\beta)'(\mathbf{X}\beta)\end{aligned}$$

Recall that the data $(X_i, Y_i), i = 1, \dots, n$ are generated from the model

$$\mathbf{Y} = \mathbf{X}\beta + \mathbf{e}$$

σ^2 reflects the size of the noise and $(\mathbf{X}\beta)'(\mathbf{X}\beta)$ reflects the size of the signal. This gives us the three golden rule of choosing λ . We should increase λ if

1. k is large
2. The signal is weak. ($\iff$ The noise is large.)

Ridge Regression

Ridge Regression

The shrinkage estimator defined in the previous section shrinks the estimates without regard to the given data. Clearly, there should be better ways to regulate the variance of $\hat{\beta}$ in a data-dependent way. Recall that the problem of large variance results from the eigenvalues of the design matrix $\mathbf{X}'\mathbf{X}$ being too small. We may fix this problem by artificially increasing the eigenvalues by adding λI_k to the design matrix, where λ is positive. Let the spectral representation of the design matrix be given by $\mathbf{X}'\mathbf{X} = UDU'$. Note that

$$\begin{aligned}\mathbf{X}'\mathbf{X} + \lambda I_k &= UDU' + \lambda UU' \\ &= U(D + \lambda I_k)U'\end{aligned}$$

Since $D + \lambda I_k$ is diagonal, the eigenvalues of the modified design matrix are given by $\lambda_i + \lambda, i = 1, \dots, n$.

Ridge Regression

The regularized design matrix $\mathbf{X}'\mathbf{X} + \lambda I_k$ is always invertible and the eigenvalues are bounded below by λ . This type of regularizing a singular or near-singular matrix is sometimes referred to as *Tikhonov regularization*.

The ridge estimator is given by

$$\hat{\beta}^{ridge} = (\mathbf{X}'\mathbf{X} + \lambda I_k)^{-1} \mathbf{X}'\mathbf{Y} \quad (1)$$

for a prespecified values of λ .

The fitted value of a ridge regression is given by

$$\begin{aligned} \hat{\mathbf{Y}}^{ridge} &= \mathbf{X} \hat{\beta}^{ridge} \\ &= \mathbf{X}(\mathbf{X}'\mathbf{X} + \lambda I_k)^{-1} \mathbf{X}'\mathbf{Y} \end{aligned}$$

Ridge Regression

Ridge regression can be motivated in another direction. Suppose that our prior knowledge of the regression model dictates that the l_2 norm of the true parameters β are bounded above by a constant K . Then, we may augment this restriction to the original squared error loss minimization problem.

$$\min_{\beta \in \mathbb{R}^k} (\mathbf{Y} - \mathbf{X}\beta)'(\mathbf{Y} - \mathbf{X}\beta) \quad \text{s.t.} \quad \beta'\beta \leq K^2 \quad (2)$$

The Lagrangian for this problem is given by

$$\mathcal{L} = (\mathbf{Y} - \mathbf{X}\beta)'(\mathbf{Y} - \mathbf{X}\beta) + \lambda(\beta'\beta - K^2) \quad (3)$$

where λ is the Lagrange multiplier.

Ridge Regression

The FOC to the constrained minimization problem is given by

$$-2\mathbf{X}'\mathbf{Y} + 2\mathbf{X}'\mathbf{X}\beta + 2\lambda\beta = 0$$

and the solution is given by (1). The equation (3) also describes the Lagrangian for the unconstrained minimization problem given by

$$\min_{\beta \in \mathbb{R}^k} (\mathbf{Y} - \mathbf{X}\beta)'(\mathbf{Y} - \mathbf{X}\beta) + \lambda\beta'\beta \quad (4)$$

where the squared error loss function has been augmented by a penalty term $\lambda\beta'\beta$. The set of solutions to the constrained minimization problem (2) is the same as the set of solutions to the penalized minimization problem (4).

Ridge Regression

The bias of $\hat{\beta}^{ridge}$ is given by

$$\mathbb{E}[\hat{\beta}^{ridge}|\mathbf{X}] - \beta = ((\mathbf{X}'\mathbf{X} + \lambda I_k)^{-1}\mathbf{X}'\mathbf{X} - I_k) \beta$$

The variance of $\hat{\beta}^{ridge}$ is given by

$$\begin{aligned}\text{Var}(\hat{\beta}^{ridge}|\mathbf{X}) &= (\mathbf{X}'\mathbf{X} + \lambda I_k)^{-1}\mathbf{X}'\text{Var}(\mathbf{Y}|\mathbf{X})\mathbf{X}(\mathbf{X}'\mathbf{X} + \lambda I_k)^{-1} \\ &= \sigma^2(\mathbf{X}'\mathbf{X} + \lambda I_k)^{-1}\mathbf{X}'\mathbf{X}(\mathbf{X}'\mathbf{X} + \lambda I_k)^{-1}\end{aligned}$$

Ridge Regression

Theorem

Suppose Assumptions 1, 2, and 3 of the classical linear regression model hold. Then, for any values $\lambda \in (0, 2\sigma^2/(\beta'\beta))$ and for any nonzero positive semi-definite weighting matrix W , the weighted MSE of $\hat{\beta}^{ridge}$ is smaller than the weighted MSE of $\hat{\beta}$.¹

Thus, as with the naive shrinkage estimator, the ridge regressor can always do better than the OLS regressor in the MSE sense, if we choose λ small enough.

¹For those interested, the proof is given in the Appendix of the Lecture notes.

Ridge Regression

Note that we can decompose $\mathbf{Y}$ into

$$\begin{aligned}\mathbf{Y} &= P_{\mathbf{X}}\mathbf{Y} + (I_n - P_{\mathbf{X}})\mathbf{Y} \\ &= \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y} + (I_n - P_{\mathbf{X}})\mathbf{Y} \\ &= \mathbf{X}\hat{\beta} + \hat{\mathbf{e}}\end{aligned}$$

The objective function (squared error) can be rewritten as

$$\begin{aligned}\frac{1}{n}(\mathbf{Y} - \mathbf{X}\beta)'(\mathbf{Y} - \mathbf{X}\beta) &= \frac{1}{n}(\mathbf{X}\hat{\beta} + \hat{\mathbf{e}} - \mathbf{X}\beta)'(\mathbf{X}\hat{\beta} + \hat{\mathbf{e}} - \mathbf{X}\beta) \\ &= \frac{1}{n}(-\mathbf{X}(\beta - \hat{\beta}) + \hat{\mathbf{e}})'(-\mathbf{X}(\beta - \hat{\beta}) + \hat{\mathbf{e}}) \\ &= \frac{1}{n}(\beta - \hat{\beta})'\mathbf{X}'\mathbf{X}(\beta - \hat{\beta}) + \hat{\mathbf{e}}'\hat{\mathbf{e}}\end{aligned}$$

where $\hat{\mathbf{e}}'\hat{\mathbf{e}}$ does not depend on β .

Ridge Regression

Thus, (2) and (4) can be rewritten as

$$\min_{\beta \in \mathbb{R}^k} \frac{1}{n} (\beta - \hat{\beta})' \mathbf{X}' \mathbf{X} (\beta - \hat{\beta}) \quad \text{s.t.} \quad \beta' \beta \leq K^2 \quad (5)$$

and

$$\min_{\beta \in \mathbb{R}^k} \frac{1}{n} (\beta - \hat{\beta})' \mathbf{X}' \mathbf{X} (\beta - \hat{\beta}) + \lambda \beta' \beta \quad (6)$$

Notice that using spectral representation $\mathbf{X}' \mathbf{X} = U D U'$, we may write

$$\begin{aligned} (\beta - \hat{\beta})' \mathbf{X}' \mathbf{X} (\beta - \hat{\beta}) &= (\beta - \hat{\beta})' U D U' (\beta - \hat{\beta}) \\ &= (U' (\beta - \hat{\beta}))' D (U' (\beta - \hat{\beta})) \\ &= (\gamma - \hat{\gamma})' D (\gamma - \hat{\gamma}) \\ &= \sum_{i=1}^k \lambda_i (\gamma_i - \hat{\gamma}_i)^2 \end{aligned}$$

where $\gamma = U' \beta$ with $\gamma_i = u_i' \beta$ and u_i is i -th column of U .

Ridge Regression

Thus, the contour set of the quadratic form $(\beta - \hat{\beta})' \mathbf{X}' \mathbf{X} (\beta - \hat{\beta})$ is given by

$$\begin{aligned} & \left\{ \beta \in \mathbb{R}^k : (\beta - \hat{\beta})' \mathbf{X}' \mathbf{X} (\beta - \hat{\beta}) = c \right\} \\ &= \left\{ \beta \in \mathbb{R}^k : \sum_{i=1}^k \lambda_i (\gamma_i - \hat{\gamma}_i)^2 = c, \gamma_i = u_i' \beta, \hat{\gamma}_i = u_i' \hat{\beta} \right\} \end{aligned}$$

Recall that the contour set $\left\{ (\beta_1, \beta_2) : \frac{\beta_1^2}{c_1} + \frac{\beta_2^2}{c_2} = c \right\}$ is given by an ellipse. The contour set of $\{(\beta_1, \beta_2) : \beta' A \beta = c, A = A'\}$ is given by a rotated ellipse. The rotation reflects the change of basis performed by the orthogonal matrix U .

Ridge Regression

Suppose $\beta \in \mathbb{R}^2$.

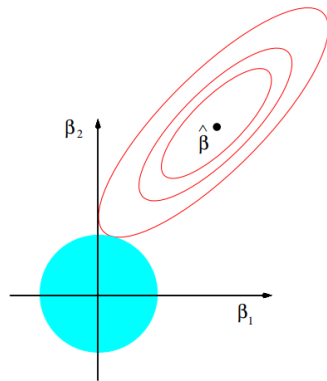


Figure 1: Ridge regression

Best Subset Selection

Best Subset Selection

Suppose that we are given p number of variables $X \in \mathbb{R}^p$ and our goal is to predict $Y \in \mathbb{R}$. Also suppose that while p is large, our prior knowledge of the problem lets us know that only $k \ll p$ number of variables are useful in predicting Y . Then, our objective function will be given by

$$\min_{\beta \in \mathbb{R}^p} (\mathbf{Y} - \mathbf{X}\beta)'(\mathbf{Y} - \mathbf{X}\beta) \quad \text{s.t.} \quad \|\beta\|_0 \leq k$$

where the l_0 norm $\|\beta\|_0$ gives the total number of nonzero elements in the vector β . In other words, among the p variables, we want to find the subset of size k such that the distance between $\mathbf{Y}$ and $\mathbf{X}\beta$ is minimized. We want to project $\mathbf{Y}$ into space spanned by only k number of columns of $\mathbf{X}$, while also finding which k columns to use so that the distance is minimized. This is NOT a convex optimization problem.

Best Subset Selection

What we can do is to fit all different possible combinations of k variables among p variables. The number of k -combinations from a given set of p variables is given by $\binom{p}{k}$. If $p = 100$, $k = 5$, the number of models that we have to fit is $7.52e + 7$. If $p = 100$, $k = 10$, we have $\binom{p}{k} = 1.73e + 13$. Choosing the best k subset among p quickly becomes impossible for a moderate size of p and k .

Forward Stepwise Selection

Forward selection is a stepwise model selection method that starts with an empty set of predictor variables and gradually builds the model by adding one variable at a time. The procedure is as follows:

1. **Initialization:**

Start with an empty set of predictor variables (e.g., no predictors in the model).

$$Y = c + e$$

2. **Addition of Variables:**

Add each individual predictor one at a time to the initial model and compute the R^2 (or MSE).

$$Y = c + X_j\beta_j + e, j = 1, \dots, p$$

Select the predictor variable that gives the largest R^2 or smallest MSE. Add the selected predictor to the model.

Forward Stepwise Selection

3. Iteration:

Given the previous model, again add each individual predictor one at a time and compute the R^2 (or MSE). Select the predictor variable that most improves the fit.

4. Stopping:

Stop if k number of variables have been selected or if the improvement of the fit due to the addition of another variable does not exceed a given threshold.

Forward selection is a *greedy algorithm*, meaning it makes locally optimal choices at each step without considering the global situation. This method is computationally efficient compared to exhaustively trying all possible variable combinations, but it may not always yield the globally optimal model.

Backward Stepwise Selection

Backward selection is another stepwise model selection method that starts with a model containing all predictor variables and gradually eliminates one variable at a time. The procedure is as follows:

- 1. Initialization:**

Start with a model that includes all predictor variables.

$$Y = c + X\beta + e$$

where $X \in \mathbb{R}^p, \beta \in \mathbb{R}^p$

- 2. Elimination of Variables:**

Eliminate each predictor one at a time and compute the R^2 (or MSE).

$$Y = c + X_{-j}\beta_{-j} + e, j = 1, \dots, p$$

where $X_{-j} \in \mathbb{R}^{p-1}, \beta_{-j} \in \mathbb{R}^{p-1}$ is $X \in \mathbb{R}^p, \beta \in \mathbb{R}^p$ without the j -th element. Eliminate the predictor variable that gives the smallest decrease in R^2 or the smallest increase in MSE.

Backward Stepwise Selection

3. Iteration:

Given the previous model, again eliminate each individual predictor one at a time and compute the R^2 (or MSE). Select the predictor variable that gives the smallest decrease to the fit.

4. Stopping:

Stop if only k number of variables are left or if the deterioration of the fit due to the elimination of an additional variable exceeds a given threshold.

Similar to forward selection, backward selection does not guarantee the best model, and its performance depends on the characteristics of the dataset. As with forward selection, it helps address issues of multicollinearity and overfitting by systematically removing less informative variables.

LASSO

References

- ▶ Trevor Hastie, Robert Tibshirani, Jerome Friedman (2009), "The Elements of Statistical Learning (ESL)"
- ▶ Lecture notes for "Convex Optimization" (10-725) by Ryan Tibshirani
- ▶ S. Boyd, Lecture notes for EE 264B, Stanford University, Spring 2010-2011

LASSO

LASSO (Least Absolute Shrinkage and Selection Operator) is a convex relaxation of l_0 norm in the best subset selection. LASSO solves the problem of

$$\min_{\beta \in \mathbb{R}^k} (\mathbf{Y} - \mathbf{X}\beta)'(\mathbf{Y} - \mathbf{X}\beta) \quad \text{s.t.} \quad \|\beta\|_1 \leq K$$

where $\|\beta\|_1 = \sum_{j=1}^k |\beta_j|$. Although LASSO does not have a closed-form solution, due to the non-differentiability of $|\beta_j|$, the solution to LASSO is characterized by what is known as **subgradient optimality condition**.

Convex Optimization

Consider general minimization problem:

$$\begin{aligned} \min_x \quad & f(x) \\ \text{s.t.} \quad & h_i(x) \leq 0, \quad i = 1, \dots, m \\ & l_j(x) = 0, \quad i = 1, \dots, r \end{aligned}$$

The problem above is called a convex optimization if

1. $f : \mathbb{R}^k \rightarrow \mathbb{R}$ is a convex function.
2. $h_i : \mathbb{R}^k \rightarrow \mathbb{R}$ is a convex function for all $i = 1, \dots, m$.
3. $l_j : \mathbb{R}^k \rightarrow \mathbb{R}$ is an affine function for all $j = 1, \dots, r$.

First-order approximation

Suppose we have $f : \mathbb{R} \rightarrow \mathbb{R}$. Then, the first order approximation of f at a point x_0 is given by

$$f(x) \approx f(x_0) + f'(x_0)(x - x_0)$$

Example: First order approximation of $f(x) = x^2$ at point x_0 would give $x_0^2 + 2x_0(x - x_0)$. Generalization of this result to $f : \mathbb{R}^k \rightarrow \mathbb{R}$ is given by

$$f(x) \approx f(x_0) + \nabla f(x_0)'(x - x_0)$$

Example: First order approximation of $f(x_1, x_2) = x_1^2 + x_2$ at (x_1^*, x_2^*) would be

$$(x_1^*)^2 + x_2^* + \begin{pmatrix} 2x_1^* & 1 \end{pmatrix} \begin{pmatrix} x_1 - x_1^* \\ x_2 - x_2^* \end{pmatrix} = (x_1^*)^2 + x_2^* + 2x_1^*(x_1 - x_1^*) + (x_2 - x_2^*)$$

Subgradient Optimality Condition

For convex and differentiable $f : \mathbb{R}^k \rightarrow \mathbb{R}$, we have

$$f(x) \geq f(x_0) + \nabla f(x_0)'(x - x_0) \quad \text{for all } x, x_0 \in \mathbb{R}^k$$

i.e., linear approximation at any point $x_0 \in \mathbb{R}^k$ always underestimates f .

A **subgradient** of a function f at $x_0 \in \mathbb{R}^k$ is any $y \in \mathbb{R}^k$ such that

$$f(x) \geq f(x_0) + y'(x - x_0) \quad \text{for all } x \in \mathbb{R}^k$$

Subgradient always exists for convex functions but may not exist for a non-convex function.

Subgradient Optimality Condition

Subdifferential of f at x_0 is the set of all subgradients of a convex function f at x_0 :

$$\begin{aligned}\partial f(x_0) &= \{y \in \mathbb{R}^k : y \text{ is a subgradient of } f \text{ at } x_0\} \\ &= \{y \in \mathbb{R}^k : f(x) \geq f(x_0) + y'(x - x_0) \quad \forall x \in \mathbb{R}^k\}\end{aligned}$$

- ▶ Subdifferential is nonempty for convex f
- ▶ $\partial f(x)$ is always closed and convex (even for nonconvex f)
- ▶ If f is differentiable at x , then $\partial f(x) = \{\nabla f(x)\}$, a singleton set.
- ▶ If $\partial f(x) = \{g\}$, then f is differentiable at x and $\nabla f(x) = g$

Subgradient Calculus

We'll use the following basic rules for subdifferential:

1. Scaling: If $a > 0$ and f convex,

$$\partial(af) = a \cdot \partial f$$

with the usual definition of a scalar multiplication of a set, i.e.,

$$c \cdot A = \{b : b = ca, a \in A\}$$

2. Addition: If f_1 and f_2 convex,

$$\partial(f_1 + f_2) = \partial f_1 + \partial f_2$$

with the usual definition of set addition. In other words,

$$A + B = \{c : c = a + b, a \in A, b \in B\}$$

Subgradient Optimality Condition

For any f (convex or not), x^* is a global minimizer if and only if 0 is in the subgradient of f at x^* . Here, we do not need to assume convexity or differentiability on f . This is called **subgradient optimality condition**.

$$f(x^*) = \min_x f(x) \Leftrightarrow 0 \in \partial f(x^*)$$

Proof.

x^* minimizes f globally if and only if $f(x) \geq f(x^*)$ for all x , which is the definition of $y = 0$ being an element of $\partial f(x^*)$:

$$f(x) \geq f(x^*) + 0'(x - x^*) = f(x^*)$$

for all x . □

For convex and differentiable functions, the subgradient optimality condition becomes the first-order condition for optimality.

LASSO

Consider the penalized formulation of the LASSO problem:

$$\min_{\beta \in \mathbb{R}^k} \frac{1}{2}(\mathbf{Y} - \mathbf{X}\beta)'(\mathbf{Y} - \mathbf{X}\beta) + \lambda \|\beta\|_1 \quad (7)$$

Let $\mathcal{L}(\beta) = \frac{1}{2}(\mathbf{Y} - \mathbf{X}\beta)'(\mathbf{Y} - \mathbf{X}\beta) + \lambda \|\beta\|_1$. Due to the subgradient optimality condition, β^* is the global minimizer of (7) if and only if $0 \in \partial \mathcal{L}(\beta^*)$. The subdifferential of $\mathcal{L}$ is given by

$$\begin{aligned} \partial \mathcal{L}(\beta) &= \partial \left(\frac{1}{2}(\mathbf{Y} - \mathbf{X}\beta)'(\mathbf{Y} - \mathbf{X}\beta) \right) + \partial (\lambda \|\beta\|_1) \\ &= \{-\mathbf{X}'(\mathbf{Y} - \mathbf{X}\beta)\} + \lambda \partial (\|\beta\|_1) \end{aligned}$$

Note that $\|\beta\|_1 = \sum_{j=1}^k |\beta_j|$. Thus, we have

$$\partial(|\beta_j|) = \begin{cases} \{\text{sign}(\beta_j)\} & \text{if } \beta_j \neq 0 \\ [-1, 1] & \text{if } \beta_j = 0 \end{cases} \implies \partial(\|\beta\|_1) = \begin{pmatrix} \partial(|\beta_1|) \\ \vdots \\ \partial(|\beta_k|) \end{pmatrix}$$

LASSO

Then, $0 \in \partial \mathcal{L}(\beta^*)$ if and only if

$$\begin{aligned} 0 &\in \{-\mathbf{X}'(\mathbf{Y} - \mathbf{X}\beta^*)\} + \lambda \partial (\|\beta^*\|_1) \\ &\iff \mathbf{X}'(\mathbf{Y} - \mathbf{X}\beta^*) = \lambda \nu \end{aligned} \tag{8}$$

for some $\nu \in \partial (\|\beta^*\|_1)$, i.e., j -th element of ν is given by

$$\nu_j = \begin{cases} \{\text{sign}(\beta_j^*)\} & \text{if } \beta_j^* \neq 0 \\ [-1, 1] & \text{if } \beta_j^* = 0 \end{cases}$$

Let x_i be the i -th column of $\mathbf{X}$. Then, (8) can be written as

$$\begin{cases} x_i'(\mathbf{Y} - \mathbf{X}\beta^*) = \lambda \cdot \text{sign}(\beta_i^*) & \text{if } \beta_i^* \neq 0 \\ |x_i'(\mathbf{Y} - \mathbf{X}\beta^*)| \leq \lambda & \text{if } \beta_i^* = 0 \end{cases}$$

LASSO

Let β^* be the solution to the LASSO problem. We call the set of covariates associated with nonzero β^* the 'active' set. The subgradient optimality condition does not give us a closed-form solution to the LASSO problem. However, it does tell us that

1. The active set of covariates all have the same sample correlation (in absolute value) with the residuals.
2. The covariates that are not in the active set have a smaller sample correlation (in absolute value) with the residuals.

Intuition: If j -th covariate has a strictly larger sample correlation with the residual, increase the estimated β_j so that the correlation becomes smaller.

LASSO with 1 Variable

Let $k = 1$ so that we only have one covariate so that $\mathbf{X} = x_1$. Then, the subgradient optimality condition gives us:

$$\begin{aligned} 0 &\in \{-\mathbf{X}'(\mathbf{Y} - \mathbf{X}\beta^*)\} + \lambda\partial(\|\beta^*\|_1) \\ &\iff \mathbf{X}'(\mathbf{Y} - \mathbf{X}\beta^*) = \lambda\nu \end{aligned} \tag{9}$$

for some $\nu \in \partial(\|\beta^*\|_1)$. Now, ν is a scalar given by

$$\nu = \begin{cases} \{\text{sign}(\beta^*)\} & \text{if } \beta^* \neq 0 \\ [-1, 1] & \text{if } \beta^* = 0 \end{cases}$$

And (9) can be written as

$$\begin{cases} \mathbf{X}'(\mathbf{Y} - \mathbf{X}\beta^*) = \lambda & \text{if } \beta^* > 0 \\ \mathbf{X}'(\mathbf{Y} - \mathbf{X}\beta^*) = -\lambda & \text{if } \beta^* < 0 \\ |\mathbf{X}'(\mathbf{Y} - \mathbf{X}\beta^*)| \leq \lambda & \text{if } \beta^* = 0 \end{cases}$$

LASSO with 1 Variable

The equations

$$\begin{cases} \mathbf{X}'(\mathbf{Y} - \mathbf{X}\beta^*) = \lambda & \text{if } \beta^* > 0 \\ \mathbf{X}'(\mathbf{Y} - \mathbf{X}\beta^*) = -\lambda & \text{if } \beta^* < 0 \\ |\mathbf{X}'(\mathbf{Y} - \mathbf{X}\beta^*)| \leq \lambda & \text{if } \beta^* = 0 \end{cases}$$

can be rewritten as

$$\begin{cases} \beta^* = \frac{1}{\mathbf{X}'\mathbf{X}}(\mathbf{X}'\mathbf{Y} - \lambda) & \text{if } \mathbf{X}'\mathbf{Y} - \lambda > 0 \\ \beta^* = \frac{1}{\mathbf{X}'\mathbf{X}}(\mathbf{X}'\mathbf{Y} + \lambda) & \text{if } \mathbf{X}'\mathbf{Y} + \lambda < 0 \\ \beta^* = 0 & \text{if } |\mathbf{X}'\mathbf{Y}| \leq \lambda \end{cases}$$

which gives us the closed-form solution to the LASSO problem.

LASSO with Orthogonal Covariates

Recall that the subgradient optimality condition characterizes the solution β^* of the LASSO problem as

$$\begin{cases} x'_i(\mathbf{Y} - \mathbf{X}\beta^*) = \lambda \cdot \text{sign}(\beta_i^*) & \text{if } \beta_i^* \neq 0 \\ |x'_i(\mathbf{Y} - \mathbf{X}\beta^*)| \leq \lambda & \text{if } \beta_i^* = 0 \end{cases}$$

where x_i be the i -th column of $\mathbf{X}$.

If the columns of $\mathbf{X}$ are orthogonal so that $\mathbf{X}'\mathbf{X}$ is a diagonal matrix, $x'_i\mathbf{X}$ is $1 \times k$ dimensional row vector with $x'_i x_i$ on the i -th element and zero otherwise. Thus, the subgradient optimality condition is given by

$$\begin{cases} x'_i\mathbf{Y} - x'_i x_i \beta_i^* = \lambda \cdot \text{sign}(\beta_i^*) & \text{if } \beta_i^* \neq 0 \\ |x'_i\mathbf{Y} - x'_i x_i \beta_i^*| \leq \lambda & \text{if } \beta_i^* = 0 \end{cases}$$

Thus, the joint problem of minimizing the objective function w.r.t. $\beta = (\beta_1, \dots, \beta_k)'$ factorizes into the individual minimization problem.

Comparison

Estimator	Formula
Best subset (size M)	$\hat{\beta}_j \cdot I(\hat{\beta}_j \geq \hat{\beta}_{(M)})$
Ridge	$\hat{\beta}_j / (1 + \lambda)$
Lasso	$\text{sign}(\hat{\beta}_j)(\hat{\beta}_j - \lambda)_+$

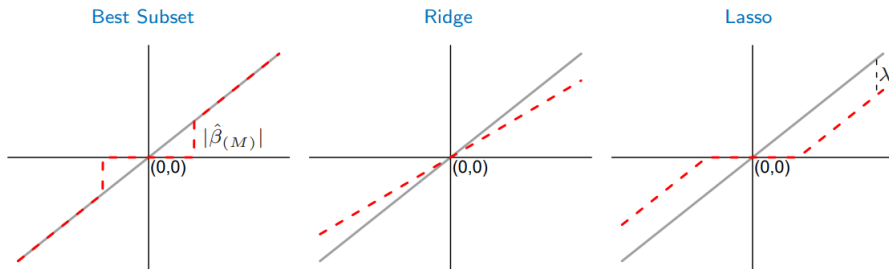


Figure 2: Figure 3.11 of ESLII

Comparison

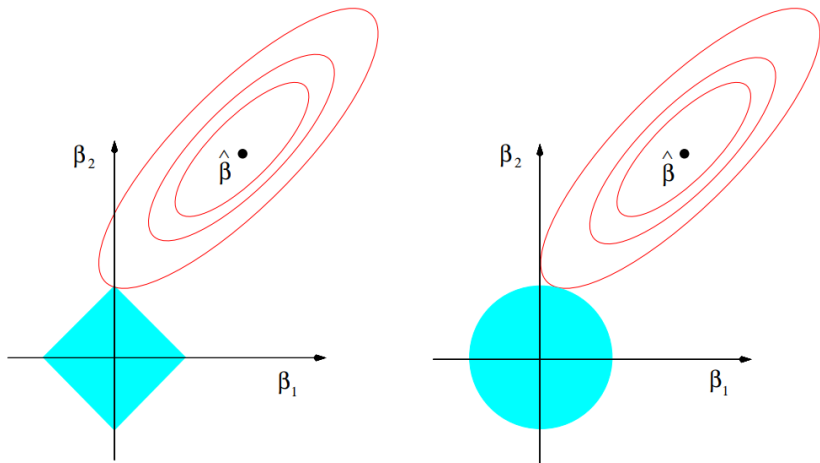


Figure 3: Figure 3.11 of ESLII

Selection of λ

In both ridge regression and LASSO, we need to fix the value of the λ before estimation. Consider the objective function used in the ridge and LASSO regression.

$$\min_{\beta \in \mathbb{R}^k} \frac{1}{2}(\mathbf{Y} - \mathbf{X}\beta)'(\mathbf{Y} - \mathbf{X}\beta) + \lambda\|\beta\|$$

where $\|\beta\|$ denotes either l_1 or l_2 norm. Just like β , we may consider using λ that minimizes the objective function. However, if we use this criterion, we'd always choose $\lambda = 0$ because norms are nonnegative. Ideally, we want to choose λ so that the out-of-sample forecast error is minimized. If we have enough training data, we may set aside a part of the data as a validation set and choose λ so that the validation error is minimized. If we don't have enough data, we may perform k -fold cross-validation. In k -fold cross-validation, the training set is divided into k -groups. We predict the values in each group using a model that has been trained on the other $k - 1$ groups and compute the sum of squared error. Then, we use the λ that minimizes this validation error.

Selection of λ

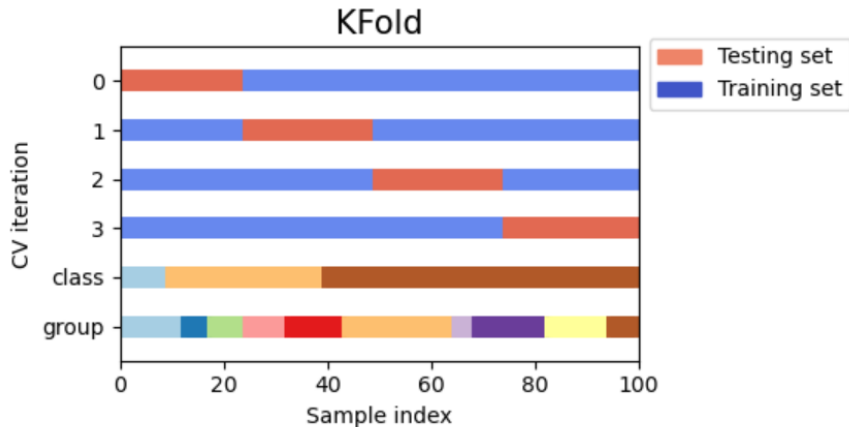


Figure 4: Figure from sklearn documentation

PCR

Singular Value Decomposition

Theorem (Singular Value Decomposition)

Let A be a $n \times m$ dimensional matrix with rank $l = \min(n, m)$. The matrix A admits an 'economy' or 'compact' singular value decomposition given as

$$A = UDV'$$

where U is $n \times l$ dimensional, V is $m \times l$ dimensional matrices with the property $U'U = V'V = I_l$, and D is $l \times l$ dimensional matrix with strictly positive diagonal elements. Furthermore, A admits a non-compact singular value decomposition given as

$$A = \tilde{U}\tilde{D}\tilde{V}'$$

where $n \times n$ dimensional matrix $\tilde{U}$ collects the eigenvectors of AA' and the $m \times m$ dimensional matrix $\tilde{V}$ collects the eigenvectors of $A'A$, and $\tilde{D}$ is a rectangular diagonal matrix with strictly positive diagonal elements.

Two Claims

Claim

For any $n \times m$ dimensional matrix A , the set of nonzero eigenvalues of $A'A$ and AA' are the same.

Proof

Suppose λ is a nonzero eigenvalue of $A'A$. By the definition of eigenvalues, there exists a nonzero $v \in \mathbb{R}^m$ such that

$$A'A v = \lambda v$$

Pre-multiplying both sides by A , we have

$$AA'A v = \lambda A v$$

Here, let $w = Av$.

Two Claims

Then, we have

$$AA'w = \lambda w$$

Thus, λ is also a nonzero eigenvalue of AA' with the associated eigenvector given by $w = Av$. Also, suppose λ is a nonzero eigenvalue of AA' . Then, we have

$$AA'v = \lambda v$$

for some nonzero $v \in \mathbb{R}^n$. Pre-multiplying both sides by A' , we have

$$A'AA'v = \lambda A'v$$

Let $w = A'v$.

Two Claims

Then, we have

$$A'Aw = \lambda w$$

Thus, λ is also a nonzero eigenvalue of $A'A$ associated with the eigenvector $w = A'v$.



Two Claims

Claim

Let $n > m$. $A'A > 0$ for any $n \times m$ matrix A of rank m .

Proof.

Note that for all nonzero $v \in \mathbb{R}^m$,

$$v'A'Av = (Av)'(Av) > 0$$

Since the rank of A is m , by the rank nullity theorem, the nullity of A is zero. This means that the only way to make $Av = 0$ is to have $v = 0$. Strict inequality holds because Av is nonzero. □

Proof of SVD

CASE 1: Suppose $n > m$

Since the rank of A is m , $A'A$ is also of rank m and is invertible. Let the spectral representation of $A'A$ be given by

$$A'A = V\Lambda V'$$

Let the diagonals of Λ be given by $\lambda_i, i = 1, \dots, m$. By the second claim above, all λ_i 's are strictly positive. Thus, $\Lambda^{1/2}$ is well-defined as a diagonal matrix with $\lambda_i^{1/2} > 0$ on the diagonals. Let U be $n \times m$ dimensional matrix given by

$$U = AV\Lambda^{-1/2}$$

We see that U is semi-orthogonal²

²An $n \times m$ dimensional matrix A with either $A'A = I_m$ or $AA' = I_n$ is called a semi-orthogonal matrix. If $n \neq m$, we cannot have both $A'A = I_m$ and $AA' = I_n$ at the same time.

Proof of SVD

because

$$\begin{aligned}U'U &= \Lambda^{-1/2}V'A'AV\Lambda^{-1/2} \\&= \Lambda^{-1/2}V'V\Lambda V'V\Lambda^{-1/2} \\&= \Lambda^{-1/2}\Lambda\Lambda^{-1/2} \\&= I_m\end{aligned}$$

Proof of SVD

Post-multiplying both sides of $AV\Lambda^{-1/2} = U$ by $\Lambda^{1/2}V'$, we have

$$A = U\Lambda^{1/2}V'$$

Let $\Lambda^{1/2} = D$. Then, we have

$$A = UDV'$$

This representation of A is the 'economy' or 'compact' singular value decomposition of A .

Proof of SVD

The graphical depiction of SVD for the case $n > m$ is given by

$$\boxed{A} = \boxed{U} \boxed{D} \boxed{V'}$$

Proof of SVD

Using the SVD of A , $A = UDV'$, we may write

$$\begin{aligned} AA' &= UDV'VD'U' \\ &= UD^2U' \\ &= U\Lambda U' \end{aligned}$$

where U is $n \times m$ dimensional semi-orthogonal matrix. This representation can be graphed as

A diagrammatic representation of the equation $AA' = U\Lambda U'$. On the left, a square box contains the expression AA' . To its right is an equals sign. Further right are three boxes: a tall vertical rectangle containing U , a small square containing Λ , and a horizontal rectangle containing U' . The boxes for U , Λ , and U' are arranged horizontally and are connected by an implied multiplication operation.

Proof of SVD

Suppose we expand the columns of U so that $\tilde{U} = (U, U_\perp)$ where $U_\perp$ is $n \times (n - m)$ dimensional matrix collecting an orthonormal basis in the orthogonal complement of U as its columns.³ Also, consider the diagonal matrix $\tilde{\Lambda}$ given by

$$\tilde{\Lambda} = \begin{pmatrix} \Lambda & 0_{m \times n-m} \\ 0_{n-m \times m} & 0_{n-m \times n-m} \end{pmatrix}$$

where the subscript below 0 denotes the dimension of the zero matrix. Then, we may write AA' as

$$AA' = U\Lambda U' = \tilde{U}\tilde{\Lambda}\tilde{U}'$$

where $\tilde{U}\tilde{U}' = I_n$. Thus, $AA' = \tilde{U}\tilde{\Lambda}\tilde{U}'$ is the spectral representation of AA' . We see that the set of nonzero eigenvalues is the same for $A'A$ and AA' , as expected.

³An orthogonal complement of a set $U \subset \mathbb{R}^n$ is the set $U^\perp = \{v \in \mathbb{R}^n : u'v = 0, \forall u \in \mathbb{R}^n\}$.

Proof of SVD

This representation can be graphed as

$$\begin{aligned} \boxed{AA'} &= \begin{array}{|c|c|} \hline U & U_{\perp} \\ \hline \end{array} \begin{array}{|c|c|} \hline \Lambda & 0 \\ \hline 0 & 0 \\ \hline \end{array} \begin{array}{|c|} \hline U' \\ \hline U'_{\perp} \\ \hline \end{array} \\ &= \begin{array}{|c|} \hline U \\ \hline \end{array} \begin{array}{|c|} \hline \Lambda \\ \hline \end{array} \begin{array}{|c|} \hline U' \\ \hline \end{array} + \begin{array}{|c|} \hline U_{\perp} \\ \hline \end{array} \begin{array}{|c|} \hline 0 \\ \hline \end{array} \begin{array}{|c|} \hline U'_{\perp} \\ \hline \end{array} \end{aligned}$$

Proof of SVD

Let $n \times m$ dimensional matrix $\tilde{D}$ be given by

$$\tilde{D} = \begin{pmatrix} D \\ 0_{n-m \times m} \end{pmatrix}$$

Then, we may derive the non-compact SVD representation given by

$$A = \tilde{U} \tilde{D} V'$$

where $\tilde{D} \tilde{D}' = \tilde{\Lambda}$. The graphical depiction of this is given as

$$\begin{array}{c} \boxed{A} = \boxed{\begin{array}{c|c} U & U_{\perp} \end{array}} \boxed{\begin{array}{c|c} D & \\ \hline 0 & \end{array}} \boxed{V'} \\ \\ = \boxed{U} \boxed{D} \boxed{V'} + \boxed{U_{\perp}} \boxed{0} \boxed{V'} \end{array}$$

Proof of SVD

CASE 2: Suppose $n < m$

Using similar methods, the 'economy' or 'compact' SVD of A is given by

$$A = UDV'$$

where now, U is $n \times n$, D is $n \times n$, and V is $m \times n$. The graphical depiction is given by

$$\boxed{A} = \boxed{U} \boxed{D} \boxed{V'}$$

Proof of SVD

The non-compact SVD representation is given by

$$A = U \tilde{D} \tilde{V}'$$

where $\tilde{D} = (D, 0_{n \times n-m})$ and $\tilde{V} = (V, V_{\perp})$. The graphical representation is given by

$$\begin{aligned} \boxed{A} &= \boxed{V} \boxed{\begin{array}{c|c} D & 0 \end{array}} \boxed{\begin{array}{c} V' \\ \hline V_{\perp}' \end{array}} \\ &= \boxed{U} \boxed{D} \boxed{V'} + \boxed{U} \boxed{0} \boxed{V_{\perp}'} \end{aligned}$$



Frobenius norm

We define the Frobenius norm of a $n \times m$ matrix A by

$$\begin{aligned}\|A\|_F &= \sqrt{\text{tr}(A'A)} \\ &= \sqrt{\sum_{i=1}^n \sum_{j=1}^m A_{ij}^2}\end{aligned}$$

If we vectorize A so that we have $nm \times 1$ column vector and compute the Euclidean norm, this is equal to the Frobenius norm.

The best low rank approximation

Theorem (The best low rank approximation)

Let A be a $n \times m$ dimensional matrix with rank $l = \min(n, m)$ with SVD

$$A = UDV'$$

Here, U is $n \times l$ dimensional, V is $m \times l$ dimensional matrices with the property $U'U = V'V = I_l$, and D is $l \times l$ dimensional matrix with strictly positive diagonal elements. Suppose we want to find a matrix B of rank $k < l$ so that the norm

$$\|A - B\|_F$$

is minimized. Then, the solution is given by

$$B = U_k D_k V_k'$$

where U_k and V_k are $n \times k$, $m \times k$ dimensional matrices collecting the first k columns of U and V . D_k is a diagonal matrix collecting the first k rows and columns of D .

Population PCA

Let $X_i \in \mathbb{R}^p$ be a random vector with the variance matrix given by

$$\Sigma = \mathbb{E}[(X_i - \mathbb{E}[X_i])(X_i - \mathbb{E}[X_i])']$$

Suppose we want to perform dimension reduction on X_i . This means that we want to find a wide matrix $B \in \mathbb{R}^{k \times p}$ ($k \ll p$) such that the low dimensional process

$$\xi_i = BX_i \in \mathbb{R}^k$$

retain as much information in the original X_i as possible. The dimension reduction technique is not complete without a way to recover the original X_i from ξ_i . We use a long matrix $C \in \mathbb{R}^{p \times k}$ to reconstruct X_i from ξ_i .

$$\tilde{X}_i = C\xi_i = CBX_i$$

Population PCA

Since $k < p$, there does not exist B and C such that $\tilde{X}_i = X_i$. The goal of PCA is to find B and C such that the 'distance' between the recovered process $\tilde{X}_i$ and the original process X_i is minimized. Let the approximation error be defined as

$$\varepsilon_i = X_i - \tilde{X}_i$$

The matrices B and C used in PCA are defined as the minimizers of the MSE defined below:

$$\begin{aligned}(B^*, C^*) &= \arg \min_{B, C} \mathbb{E}[(\varepsilon_i - \mathbb{E}[\varepsilon_i])'(\varepsilon_i - \mathbb{E}[\varepsilon_i])] \\ &= \arg \min_{B, C} \text{tr}(\mathbb{E}[(\varepsilon_i - \mathbb{E}[\varepsilon_i])(\varepsilon_i - \mathbb{E}[\varepsilon_i])']) \\ &= \arg \min_{B, C} \text{tr}(\text{Var}(\varepsilon_i))\end{aligned}\tag{10}$$

MSE minimization formulation of population PCA

Theorem

The solution (B^, C^*) to (10) is given by*

$$B^* = V_k'$$

$$C^* = V_k$$

where $p \times k$ dimensional matrix V_k collects the eigenvectors of Σ associated with k largest eigenvalues as its columns. The process (ξ_i) defined using $B^ = V_k'$ are called 'principal components' of X_i .*

Proof of Theorem

Note that we have $\varepsilon_i = (I_p - CB)X_i$, which implies

$$\begin{aligned}\text{Var}(\varepsilon_i) &= (I_p - CB)\text{Var}(X_i)(I_p - CB)' \\ &= (I_p - CB)\Sigma(I_p - CB)'\end{aligned}$$

Thus, the objective function in (10) can be rewritten as

$$\begin{aligned}\text{tr}(\text{Var}(\varepsilon_i)) &= \text{tr}((I_p - CB)\Sigma(I_p - CB)') \\ &= \text{tr}\left((I_p - CB)\Sigma^{1/2}\Sigma^{1/2}(I_p - CB)'\right) \\ &= \text{tr}\left(\Sigma^{1/2}(I_p - CB)'(I_p - CB)\Sigma^{1/2}\right) \\ &= \text{tr}\left(\left((I_p - CB)\Sigma^{1/2}\right)' \left((I_p - CB)\Sigma^{1/2}\right)\right)\end{aligned}$$

Thus, the objective function becomes the squared Frobenius norm of $(I_p - CB)\Sigma^{1/2}$.

Proof of Theorem

Notice that $CB\Sigma^{1/2}$ is of rank k . Let the spectral representation of Σ be given by

$$\Sigma = V\Lambda V'$$

By Theorem 3, the best rank k approximation to $\Sigma^{1/2}$ is given by $\Sigma_k = V_k\Lambda_k^{1/2}V_k'$. Suppose we take $B = V_k'$ and $C = V_k$. Then, we have

$$\begin{aligned}CB\Sigma^{1/2} &= V_k V_k' \Sigma^{1/2} \\&= V_k V_k' V \Lambda^{1/2} V' \\&= V_k \Lambda_k^{1/2} V_k'\end{aligned}$$

which gives us the best rank k approximation. Notice that we've used $V_k' V = (I_k, 0_{k \times p-k})$ in the last equality. Since Theorem 3 tells us that we cannot do any better than this, the solution (B^*, C^*) is given by (V_k', V_k) . □

Identification of ξ_i

Note that the solution is not unique. Let R be any invertible $k \times k$ dimensional matrix. Let $(B^*, C^*) = (RV'_k, V_k R^{-1})$. Then, we also have

$$\begin{aligned} CB\Sigma^{1/2} &= V_k R^{-1} R V'_k \Sigma^{1/2} \\ &= V_k V'_k V \Lambda^{1/2} V' \\ &= V_k \Lambda_k^{1/2} V'_k \end{aligned}$$

so that we may as well write $\xi_i = B^* X_i = R V'_k X_i$ for any $k \times k$ dimensional invertible matrix R . In this class, we'll fix $R = I_k$. This implies that ξ_i is defined in such a way that leading elements have larger variance than the following elements.

Variance of ξ_i

Notice that the variance of the principal components $\xi_i = B^* X_i = V_k' X_i$ is given by

$$\begin{aligned}\text{Var}(\xi_i) &= \text{Var}(V_k' X_i) \\ &= V_k' \text{Var}(X_i) V_k \\ &= V_k' V \Lambda V' V_k \\ &= \Lambda_k\end{aligned}$$

Thus, the principal components are by construction uncorrelated with each other. Let ξ_i be written as

$$\xi_i = \begin{pmatrix} \xi_{1i} \\ \vdots \\ \xi_{ki} \end{pmatrix}$$

Then, we have $\text{Var}(\xi_{ji}) = \lambda_j \geq \lambda_l = \text{Var}(\xi_{li})$ for $j \geq l$.

Constrained variance maximization formulation of population PCA

Theorem

Let ξ_i be given by $\xi_i = BX_i$ for some $k \times p$ dimensional matrix B . The solution to the MSE minimization problem (10) and the solution to the constrained variance maximization problem below are the same.

$$\begin{aligned} B^* &= \arg \min_B \text{tr}(\text{Var}(\xi_i)) \quad \text{s.t. } BB' = I_k \\ &= \arg \min_B \text{tr}(B\Sigma B') \quad \text{s.t. } BB' = I_k \end{aligned} \tag{11}$$

In other words, we have $B^* = V'_k$.

Proof of Theorem

Let us again consider the objective function of the MSE minimization problem (10).

$$\text{tr}(\text{Var}(\varepsilon_i)) = \text{tr}((I_p - CB)\Sigma(I_p - CB)')$$

Suppose we restrict C to satisfy $C = B'$ and $C'C = BB' = I_k$. Note that these restrictions are satisfied at the solution to (10). Then, we have

$$\begin{aligned}\text{tr}(\text{Var}(\varepsilon_i)) &= \text{tr}((I_p - B'B)\Sigma(I_p - B'B)') \\ &= \text{tr}((I_p - B'B)^2\Sigma) \\ &= \text{tr}((I_p - B'B)\Sigma) \\ &= \text{tr}(\Sigma) - \text{tr}(B\Sigma B') \\ &= \text{tr}(\Sigma) - \text{tr}(\text{Var}(\xi_i))\end{aligned}$$

where we've used the definition $\xi_i = BX_i$.

Proof of Theorem

Thus, we have

$$\text{tr}(\Sigma) = \text{tr}(\text{Var}(\xi_i)) + \text{tr}(\text{Var}(\varepsilon_i)) \quad (12)$$

This means that under the restrictions $C = B'$ and $C'C = BB' = I_k$, the trace of variance of X_i decomposes into the trace of variance of ξ_i and the trace of variance of MSE. Thus, under these restrictions, maximizing $\text{tr}(\text{Var}(\xi_i))$ and minimizing $\text{tr}(\text{Var}(\varepsilon_i))$ are equivalent. This is because Σ is a characteristic of X_i and does not involve either B or C .

Here, minimizing $\text{tr}(\text{Var}(\varepsilon_i))$ with or without constraints gives equivalent solutions. This is because the unconstrained minimization problem yields the solution $B^* = V'_k, C^* = B^{*'}$, which satisfies the constraints. Thus, the solution to (11) should also be given by $B^* = V'_k$. □

Sequential variance maximization

We motivate this equivalence in another, perhaps a more constructive, direction. Suppose we start with a sequential constrained variance maximization problem. In other words, we find a linear combination of $X_i \in \mathbb{R}^p$ so that its variance is maximized. The linear combination of X_i can be written as $v'X_i$ for some $v \in \mathbb{R}^p$, which gives the problem:

$$\max_{v \in \mathbb{R}^p} \text{Var}(v'X_i) \quad \text{s.t. } v'v = 1$$

After finding the solution v^* to the above problem, we define the random variable $\xi_{1i} = v^{*'}X_i$ as the first principal component of X_i .

Sequential variance maximization

The above is an equality-constrained problem and may be solved using the method of Lagrangian. Note that the Lagrangian is given by

$$\begin{aligned}\mathcal{L} &= \text{Var}(v'X_i) + \lambda(1 - v'v) \\ &= v'\Sigma v + \lambda(1 - v'v)\end{aligned}$$

The FOC w.r.t. v is given by

$$\begin{aligned}2\Sigma v - 2\lambda v &= 0 \\ \implies \Sigma v &= \lambda v\end{aligned}$$

Thus, the solution v^* is given by one of the eigenvectors of Σ , and the maximized value is given by $v^{*'}\Sigma v^* = v^{*'}(\lambda v^*) = \lambda^*$ is the associated eigenvalue. Since we are solving a maximization problem, the solution is given by the eigenvector v_1 associated with the largest eigenvalue. Thus, the first principal component $\xi_{1i} = v_1'X_i$ is equivalent to the first element of the principal components ξ_i defined as the solution to the MSE minimization problem (10).

Sequential variance maximization

After obtaining the solution to the first problem, suppose that we solve the next problem.

$$\max_{v \in \mathbb{R}^p} \text{Var}(v'X_i) \quad \text{s.t.} \quad \text{Cov}(v_1'X_i, v'X_i) = 0, v'v = 1$$

Notice that

$$\begin{aligned} \text{Cov}(v_1'X_i, v'X_i) &= \mathbb{E}[(v_1'X_i - \mathbb{E}(v_1'X_i))(v'X_i - \mathbb{E}(v'X_i))] \\ &= v_1'\mathbb{E}[(X_i - \mathbb{E}(X_i))(X_i - \mathbb{E}(X_i))]v \\ &= v_1'\Sigma v \\ &= (\lambda_1 v_1')v \end{aligned}$$

Thus, the above problem may be written as:

$$\max_{v \in \mathbb{R}^p} \text{Var}(v'X_i) \quad \text{s.t.} \quad v_1'v = 0, v'v = 1$$

After finding solution v^* , we define $\xi_{2i} = v^{*'}X_i$ as the second principal component of X_i .

Sequential variance maximization

The Lagrangian of the second problem is given by

$$\begin{aligned}\mathcal{L} &= \text{Var}(v'X_i) + \mu(v'_1v) + \lambda(1 - v'v) \\ &= v'\Sigma v + \mu(v'_1v) + \lambda(1 - v'v)\end{aligned}$$

The FOC is given by

$$2\Sigma v + \mu v_1 - 2\lambda v = 0$$

Pre-multiply both sides by v'_1 to obtain

$$\begin{aligned}2v'_1\Sigma v + \mu v'_1v_1 - 2\lambda v'_1v &= 0 \\ \implies 2v'_1\lambda_1v + \mu - 2\lambda v'_1v &= 0 \\ \implies 2\lambda_1v'_1v + \mu - 2\lambda v'_1v &= 0 \\ \implies \mu &= 0\end{aligned}$$

where we've used $\Sigma v_1 = \lambda_1 v_1$, $v'_1v = 0$ and $v'_1v_1 = 1$. Thus the Lagrange multiplier μ is zero.

Sequential variance maximization

This leads to the FOC

$$\begin{aligned}2\Sigma v - 2\lambda v &= 0 \\ \implies \Sigma v &= \lambda v\end{aligned}$$

which again defines eigenvectors of Σ . Since $v_1'v = 0$, the eigenvector v_2 associated with the second largest eigenvalue is the solution. Thus, the second principal component $\xi_{2i} = v_2'X_i$ is equivalent to the second element of the principal components ξ_i defined as the solution to the MSE minimization problem (10). We can proceed this way to show that the constrained variance maximization problem

$$\begin{aligned}\max_{v \in \mathbb{R}^p} \text{Var}(v'X_i) \quad \text{s.t. } v_l'v &= 0, \quad l = 1, \dots, j-1 \\ v'v &= 1\end{aligned}$$

has the solution given by the eigenvector v_j associated with the j -th largest eigenvalue.

Explained Variance

Let us rewrite (12) here for convenience.

$$\text{tr}(\Sigma) = \text{tr}(\text{Var}(\xi_i)) + \text{tr}(\text{Var}(\varepsilon_i))$$

Evaluating (12) at the solution $B^* = V_k'$, we have

$$\begin{aligned}\text{tr}(\Sigma) &= \text{tr}(B^* \Sigma B^{*'}) + \text{tr}(\text{Var}(\varepsilon_i)) \\ &= \text{tr}(V_k' \Sigma V_k) + \text{tr}(\text{Var}(\varepsilon_i)) \\ &= \text{tr}(V_k' V \Lambda V' V_k) + \text{tr}(\text{Var}(\varepsilon_i)) \\ &= \text{tr}(\Lambda_k) + \text{tr}(\text{Var}(\varepsilon_i))\end{aligned}$$

Thus,

$$\text{Explained Variance} = \frac{\text{tr}(\Lambda_k)}{\text{tr}(\Sigma)} = \frac{\text{tr}(\Lambda_k)}{\text{tr}(\Lambda)}$$

denotes the proportion of variance of X_i that is explained by ξ_i . This is strictly increasing in k and becomes 1 if $k = p$.

Sample PCA

We've seen that the construction of principal components ξ_i involves the original random vector X_i and the eigenvectors of the variance matrix Σ . In practice, we do not know the variance matrix Σ and only have observations $X_i \in \mathbb{R}^p, i = 1, \dots, n$. Thus, we first estimate the variance matrix using a sample analog estimator⁴

$$\hat{\Sigma} = \frac{1}{n} \sum_{i=1}^n X_i X_i'$$

and estimate the principal components by

$$\hat{\xi}_i = \hat{V}_k' X_i \tag{13}$$

where $\hat{V}_k$ collects the leading k eigenvectors of $\hat{\Sigma}$ as its columns.

⁴For notational simplicity, let us assume that the observations X_i have been demeaned using its sample mean.

Sample PCA

Here, $\hat{\xi}_i \in \mathbb{R}^k$ denotes the low dimensional process from which we can recover X_i using

$$\tilde{X}_i = \hat{V}_k \hat{\xi}_i = \hat{V}_k \hat{V}_k' X_i$$

Sample PCA

Let $\mathbf{X} \in \mathbb{R}^{n \times p}$ be the matrix collecting i -th sample X_i' at i -th row. Then, we have

$$\hat{\Sigma} = \frac{1}{n} \sum_{i=1}^n X_i X_i' = \frac{1}{n} \mathbf{X}' \mathbf{X}$$

Let the spectral representation of $\hat{\Sigma} = \frac{1}{n} \mathbf{X}' \mathbf{X}$ be given by

$$\begin{aligned} \frac{1}{n} \mathbf{X}' \mathbf{X} &= \frac{1}{n} \hat{V} \hat{\Lambda} \hat{V}' \\ &= \frac{1}{n} \sum_{j=1}^p \hat{\lambda}_j \hat{v}_j \hat{v}_j' \\ &= \frac{1}{n} \sum_{j=1}^k \hat{\lambda}_j \hat{v}_j \hat{v}_j' + \frac{1}{n} \sum_{j=k+1}^p \hat{\lambda}_j \hat{v}_j \hat{v}_j' \end{aligned}$$

Sample PCA

The sample PCA can be thought of decomposition of $\hat{\Sigma} = \frac{1}{n}\mathbf{X}'\mathbf{X}$ into two parts. The first part

$$\frac{1}{n} \sum_{j=1}^k \hat{\lambda}_j \hat{v}_j \hat{v}_j' = \frac{1}{n} \hat{V}_k \hat{\Lambda}_k \hat{V}_k'$$

denotes the sample variance of the low-rank representation $\tilde{X}_i = \hat{V}_k \hat{V}_k' X_i$. The second part

$$\frac{1}{n} \sum_{j=k+1}^p \hat{\lambda}_j \hat{v}_j \hat{v}_j'$$

denotes the sample variance of the approximation error.

Sample PCA (Sequential variance max)

Motivation 1:

Consider the problem of finding a linear combination of columns of $\mathbf{X}$ so that the Euclidean norm of the linear combination is maximized. In other words, we solve

$$\max_{v \in \mathbb{R}^p} (\mathbf{X}v)'(\mathbf{X}v) \quad \text{s.t. } v'v = 1$$

We may readily show that the solution is given by the eigenvector v_1 of $\frac{1}{n}\mathbf{X}'\mathbf{X}$ associated with the largest eigenvalue.

Sample PCA (Sequential variance max)

Subsequently, suppose that we solve the problem below.

$$\max_{v \in \mathbb{R}^p} \text{var}(\mathbf{X}v)'(\mathbf{X}v) \quad \text{s.t. } v_1'v = 0, v'v = 1$$

Then, the solution is given by the eigenvector v_2 of $\frac{1}{n}\mathbf{X}'\mathbf{X}$ associated with the second largest eigenvalue.

Sample PCA (Sequential variance max)

Note that we have

$$\mathbf{X}v_1 = \begin{pmatrix} X'_1 v_1 \\ \vdots \\ X'_n v_1 \end{pmatrix} = \begin{pmatrix} \hat{\xi}_{11} \\ \vdots \\ \hat{\xi}_{1n} \end{pmatrix}$$

where $\hat{\xi}_{1i}$ denotes the first element of $\hat{\xi}_i$. Thus, by construction, $\hat{\xi}_{1i}$ has the largest variation possible along the rows, i.e., the sample variance of $\hat{\xi}_{1i}$ is the largest possible.

Sample PCA (SVD)

Motivation 2:

Let $l = \min(n, p)$. Let the economy SVD of $n \times p$ dimensional matrix $\mathbf{X}$ be given by

$$\mathbf{X} = \hat{U} \hat{D} \hat{V}'$$

where $\hat{U}$ is $n \times l$ dimensional, $\hat{V}$ is $p \times l$ dimensional matrices with the property $\hat{U}'\hat{U} = \hat{V}'\hat{V} = I_l$, and $\hat{D}$ is $l \times l$ dimensional matrix with strictly positive diagonal elements. Using Theorem 3, the best rank k approximation of $\mathbf{X}$ is given by

$$\tilde{\mathbf{X}} = \hat{U}_k \hat{D}_k \hat{V}_k'$$

Since the right singular vectors of $\mathbf{X}$ are equal to the eigenvectors of $\mathbf{X}'\mathbf{X}$, $\hat{V}_k$ in the above equation is actually the same as $\hat{V}_k$ in (13).

Sample PCA (SVD)

By post-multiplying $\hat{V}_k$ to both sides, we have

$$\tilde{\mathbf{X}}\hat{V}_k = \hat{U}_k\hat{D}_k$$

Consider the LHS:

$$\tilde{\mathbf{X}}\hat{V}_k = \mathbf{X}\hat{V}_k = \begin{pmatrix} X'_1\hat{V}_k \\ \vdots \\ X'_n\hat{V}_k \end{pmatrix} = \begin{pmatrix} \hat{\xi}'_1 \\ \vdots \\ \hat{\xi}'_n \end{pmatrix}$$

Thus, $\tilde{\mathbf{X}}\hat{V}_k = \hat{U}_k\hat{D}_k$ collects the samples of estimated $\hat{\xi}_i$ on each of its rows. Since the singular values are always strictly positive and ordered in decreasing values, we see that the first column of $\tilde{\mathbf{X}}\hat{V}_k = \hat{U}_k\hat{D}_k$ must have the largest Euclidean norm, as expected.

Sample PCA

For later use, we introduce the notation for the matrix collecting samples of estimated $\hat{\xi}_i$ here.

$$\hat{\mathbf{F}} = \begin{pmatrix} \hat{\xi}'_1 \\ \vdots \\ \hat{\xi}'_n \end{pmatrix}$$

Change of Basis

Sample PCA can also be thought of as a change of basis from the elementary basis $(e_i)_{i=1}^n$ to the principal component basis $(\hat{v}_i)_{i=1}^n$. Consider the basis expansion of i -th sample X_i using the principal component basis.

$$X_i = \sum_{j=1}^p (\hat{v}_j' X_i) \hat{v}_j$$

After fixing the basis $(\hat{v}_i)$, the vector $V'X_i = (\hat{v}_1'X_i, \hat{v}_2'X_i, \dots, \hat{v}_p'X_i)' \in \mathbb{R}^p$ contains all the information on X_i . The transformation $\mathbf{X} \mapsto \mathbf{X}\hat{V}$ implies the change of basis from $(e_i)_{i=1}^n$ to $(\hat{v}_i)_{i=1}^n$.

Change of Basis

By construction, $\sum_{i=1}^n (\hat{v}_j' X_i)^2$ is largest for $j = 1$ and is decreasing in j . If we want to approximate X_i using only k number of basis, it would be best to use the approximation given by

$$X_i \approx \sum_{j=1}^k (\hat{v}_j' X_i) \hat{v}_j$$

in the sense that the Euclidean norm of the approximation error is minimized. This approximation is written as $\mathbf{X} \mapsto \mathbf{X} \hat{V}_k$.

Choosing k

As with ridge regression and LASSO, we may use cross-validation to choose the value k . Another commonly used practice is to look at the explained variance. In the last section, the explained variance was given as

$$\text{Explained Variance} = \frac{\text{tr}(\Lambda_k)}{\text{tr}(\Sigma)} = \frac{\text{tr}(\Lambda_k)}{\text{tr}(\Lambda)}$$

The explained (sample) variance is given by

$$\text{Explained (Sample) Variance} = \frac{\text{tr}(\hat{\Lambda}_k)}{\text{tr}(\hat{\Sigma})} = \frac{\text{tr}(\hat{\Lambda}_k)}{\text{tr}(\hat{\Lambda})}$$

To choose k , we first calculate the explained variance for each value of k . Increasing k will increase the explained variance. However, the increase of the explained variance will typically decrease as we increase k . We use the value k^* such that the increase of explained variance caused by an increase of k is below a certain threshold.

PCR

Finally, we are ready to talk about the *principal components regression*. PCR is a three-step procedure.

Step 1: Perform sample PCA on $\mathbf{X}$ to get the estimated factors

$$\hat{\mathbf{F}} = \mathbf{X}\hat{V}_k$$

Step 2: Regress $\mathbf{Y}$ on $\hat{\mathbf{F}}$ to get $k \times 1$ dimensional coefficient vector

$$\hat{\gamma}^{PCR} = (\hat{\mathbf{F}}'\hat{\mathbf{F}})^{-1}\hat{\mathbf{F}}'\mathbf{Y}$$

Step 3: Change the basis to the original basis using

$$\hat{\beta}^{PCR} = \hat{V}_k\hat{\gamma}^{PCR}$$

Then, the fitted value will be given by

$$\hat{\mathbf{Y}}^{PCR} = \hat{\mathbf{X}}\hat{\beta}^{PCR}$$

Comparison with other methods

The estimated OLS coefficients are given by

$$\hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y}$$

The estimated ridge coefficients are given by

$$\hat{\beta}^{ridge} = (\mathbf{X}'\mathbf{X} + \lambda I_p)^{-1}\mathbf{X}'\mathbf{Y}$$

Let the singular value decomposition of $\mathbf{X}$ be given by

$$\mathbf{X} = \mathbf{U}\mathbf{D}\mathbf{V}'$$

Comparison with other methods

Then, the estimated OLS coefficients can be written as

$$\begin{aligned}\hat{\beta} &= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y} \\ &= ((UDV')'UDV')^{-1}(UDV')'\mathbf{Y} \\ &= (VD^2V')^{-1}VDU'\mathbf{Y} \\ &= (VD^{-2}V')VDU'\mathbf{Y} \\ &= VD^{-1}U'\mathbf{Y} \\ &= \left(\sum_{i=1}^p \frac{1}{\delta_i} v_i u_i' \right) \mathbf{Y} \\ &= \sum_{i=1}^p \frac{1}{\delta_i} (u_i' \mathbf{Y}) v_i\end{aligned}$$

Comparison with other methods

the estimated ridge coefficients can be written as

$$\begin{aligned}\hat{\beta}^{ridge} &= ((UDV')'UDV' + \lambda I_p)^{-1}(UDV')'\mathbf{Y} \\ &= (VD^2V' + \lambda I_p)^{-1}(UDV')'\mathbf{Y} \\ &= (VD^2V' + \lambda VV')^{-1}(UDV')'\mathbf{Y} \\ &= V(D^2 + \lambda I_n)^{-1}V'(UDV')'\mathbf{Y} \\ &= V(D^2 + \lambda I_n)^{-1}DU'\mathbf{Y} \\ &= \left(\sum_{i=1}^p \frac{\delta_i}{\delta_i^2 + \lambda} v_i u_i' \right) \mathbf{Y} \\ &= \sum_{i=1}^p \frac{\delta_i}{\delta_i^2 + \lambda} (u_i' \mathbf{Y}) v_i\end{aligned}$$

Comparison with other methods

The estimated PCR coefficients are given as

$$\begin{aligned}\hat{\beta}^{PCR} &= V_k \hat{\gamma}^{PCR} \\ &= V_k (\hat{\mathbf{F}}' \hat{\mathbf{F}})^{-1} \hat{\mathbf{F}}' \mathbf{Y} \\ &= V_k ((\mathbf{X} V_k)' \mathbf{X} V_k)^{-1} (\mathbf{X} V_k)' \mathbf{Y} \\ &= V_k ((U D V' V_k)' U D V' V_k)^{-1} (U D V' V_k)' \mathbf{Y} \\ &= V_k ((U_k D_k)' U_k D_k)^{-1} (U_k D_k)' \mathbf{Y} \\ &= V_k D_k^{-2} (U_k D_k)' \mathbf{Y} \\ &= V_k D_k^{-1} U_k' \mathbf{Y} \\ &= \sum_{i=1}^k \frac{1}{\delta_i} (u_i' \mathbf{Y}) v_i\end{aligned}$$

Comparison with other methods

We see that all three methods estimate β as a linear combination of the eigenvectors (v_i). In $\hat{\beta}^{ridge}$, the eigenvector associated with small δ_i is regularized by multiplying by $\frac{\delta_i^2}{\delta_i^2 + \lambda}$ to each of $\frac{1}{\delta_i} (u_i' \mathbf{Y}) v_i$ in the OLS estimates. Note that if δ_i is small, $\frac{\delta_i^2}{\delta_i^2 + \lambda}$ is small. $\hat{\beta}^{PCR}$ regularizes the model by just not using the eigenvector associated with small δ_i .

Comparison with other methods

We may also show that

$$\hat{\mathbf{Y}} = \sum_{i=1}^p (u_i' \mathbf{Y}) u_i$$

$$\hat{\mathbf{Y}}^{ridge} = \sum_{i=1}^p \frac{\delta_i^2}{\delta_i^2 + \lambda} (u_i' \mathbf{Y}) u_i$$

$$\hat{\mathbf{Y}}^{PCR} = \sum_{i=1}^k (u_i' \mathbf{Y}) u_i$$

Estimation of a linear model is done by finding a linear combination of columns of $\mathbf{X}$ so that the distance between the linear combination and $\mathbf{Y}$ is minimized. Since the range of $\mathbf{X}$ and the range of U in $\mathbf{X} = UDV'$ is the same, we may instead find a linear combination of U . The above three expressions show how the (u_i) 's are linearly combined in each of the methods. Ridge regression and PCR are best interpreted in the changed basis u_i , instead of the original basis.

Comparison with other methods

On the other hand, forward (backward) stepwise regression and LASSO are best explained using the original basis. Contrary to ridge or PCR, the regularization is placed directly on columns of $\mathbf{X}$, instead of (u_i) .